

Course Plan

Semester: 7

Year: 2026-27

Course Title: Agentic AI	Course Code:26ECAC401
Total Contact hrs.: 58	Duration of ESA: 2 Hrs.
ISA Marks: 50	ESA Marks: 50
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Checked By: Dr. Narayan D.G	Date: 12-07-2026

Prerequisites: Basics of Python programming, Machine Learning and Generative AI concepts, Large Language Models (LLMs), and Software Engineering fundamentals.

Course Overview:

The course enables students to design, build, and deploy Agentic Artificial Intelligence (Agentic AI) systems that exhibit autonomy, goal-directed behavior, and adaptive decision-making. Beginning with the evolution from Generative AI to Agentic AI and the limitations of standalone LLM-based applications, the course introduces the agent lifecycle, the plan–act–observe–revise loop, and design principles for scalable, goal-driven agents. Students then explore agent frameworks such as LangChain, tool integration and function calling, reusable skills, memory management (short-term state and long-term retrieval), and the Model Context Protocol (MCP) for connecting agents to external systems. The course further develops multi-agent orchestration using agentic node graphs with LangGraph and conversational collaboration with AutoGen, covering role-based specialization, parallel execution, swarms, fault tolerance, and deep research workflows with evidence synthesis. Finally, the course addresses learning agents and responsible deployment: reinforcement learning-based alignment (RLHF, RLAIF, DPO), context engineering, self-correcting agents using Reflexion, human-in-the-loop oversight, the agentic SDLC, and safety and governance through approval gates, audit logs, and evaluation. Through hands-on labs and a discipline-specific capstone, students gain the theoretical grounding and practical skills required to design, build, and responsibly deploy next-generation autonomous AI systems.

Course Outcomes

At the end of the course the students should be able to:

1. Explain the foundations, architecture, and planning strategies of agentic AI.
2. Design agentic systems using frameworks, tools, skills, and memory (MCP).
3. Develop multi-agent orchestration using LangGraph and collaborative agents.
4. Analyse safety, governance, and alignment in autonomous agent systems.
5. Build and demonstrate an end-to-end, discipline-specific agentic AI worklet as a course project, with a technical report and presentation.

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (POs)

Course Title: Agentic AI	Semester: 7
Course Code: 26ECAC401	Year: 2026-27

Course Outcomes (COs) / Program Outcomes (POs)	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1. Explain the foundations, architecture, and planning strategies of agentic AI.	H	M												
2. Design agentic systems using frameworks, tools, skills, and memory (MCP).	H	M												
3. Develop multi-agent orchestration using LangGraph and collaborative agents.	H	M												
4. Analyse safety, governance, and alignment in autonomous agent systems	H	M												
5. Build and demonstrate an end-to-end, discipline-specific agentic AI worklet as a course project, with a technical report and presentation.	H	M	M		M				M	M				

Degree of compliance **L**: Low **M**: Medium **H**: High

Competency addressed in the Course and corresponding Performance Indicators

Competency	Performance Indicators (PIs)	Course Specific performance indicators (CSPIs)
1.4 Demonstrate competence in computer science engineering knowledge.	1.4.1 Apply knowledge of suitable data structures and/or programming paradigms to solve problems.	1.4.1.1 Apply suitable data structures like lists, dictionaries and queues to manage agent memory, context, and tool calls. 1.4.1.2 Apply object-oriented or functional programming paradigms to design agent architectures and runtimes.

		1.4.1.3 Apply event-driven programming to design agent interactions, tool calling, and MCP connectors in dynamic environments.
2.1 Demonstrate an ability to identify and characterize an engineering problem	2.1.2 Identify critical systems/subsystems, variables, and constraints that characterize the problem environment by analysing practical contexts and real-world limitations.	2.1.2.1 Identify agent goals, environmental constraints, failure modes, data needs, tools, and coordination dependencies for an Agentic AI problem.
2.2 Demonstrate an ability to apply principles of mathematics, natural sciences, and engineering to formulate and analyse complex problems, incorporating literature review to support sustainable development.	2.2.1 Decompose complex problems into interconnected sub-problems.	2.2.1.1 Decompose an agentic task into goals, sub-goals, agent roles, tool calls, decision points, and merge steps.
2.3 Demonstrate an ability to formulate and interpret a model	2.3.1 Apply computer engineering principles to formulate models	2.3.1.1 Design systems where multiple AI agents coordinate, compete, or collaborate using role-based specialization and communication. 2.3.1.2 Design LLM-based agents with tools, memory, skills, and MCP connectors. 2.3.1.5 Apply human-in-the-loop and human-agent collaboration principles to build safe, governed agentic systems.
5.3 Demonstrate an ability to identify/create modern IT tools for engineering activities.	5.3.1 Demonstrate proficiency in using IT tools for engineering activities	5.3.1.1 Demonstrate proficiency in using LangChain and LangGraph to build and orchestrate agentic workflows. 5.3.1.2 Use APIs, memory stores, and MCP connectors to extend agent capability and context. 5.3.1.3 Integrate observability, audit logging, and approval-gate tools for safe agent deployment.

10.1 Demonstrate effective use of written communication skills relevant to the engineering discipline that conveys information effectively to both technical and non-technical stake holders	10.1.3 Write technical short reports, formal e-mails, reports documenting experimental or simulation methods and results	10.1.3.1 Write detailed project and capstone reports adhering to academic and engineering standards. 10.1.3.2 Prepare structured evaluation reports documenting agent design, experiments, results, and safety analysis.
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Course Content

Course Code:26ECAC401	Course Title: Agentic AI	
L-T-P: 2-0-1	Credits:3	Contact Hrs: 4hrs/week
ISA Marks: 50	ESA Marks: 50	Total Marks: 100
Teaching Hrs: 30	Laboratory Hrs: 28	Exam Duration: 2 hrs

Content	Hrs
Unit - 1	
Chapter No 1. Foundations of Agentic AI: Evolution from Generative AI to Agentic AI systems; limitations of standalone LLM-based applications in real-world deployments; definition and characteristics of autonomous agents; agent lifecycle and the plan–act–observe–revise loop; agent architecture; goal decomposition and iterative reasoning; planning strategies; design principles for building scalable and goal-driven AI agents.	7
Chapter No 2. Agent Frameworks, Tools, Skills & Memory: Design of agentic systems using frameworks such as LangChain; tool integration and function calling including API orchestration, code execution, and external system interaction; building reusable skills; memory management with short-term state and long-term retrieval stores; context handling and optimization; Model Context Protocol (MCP) and connectors; unified agent architecture combining reasoning, memory, tools, and environment; agent runtime patterns (OpenCLA/OpenClaw) with logging for real-world deployment.	8
Unit - 2	
Chapter No 3. Multi-Agent Orchestration: Designing distributed AI systems using multi-agent architectures; role-based specialization and coordination strategies; agentic node graphs and graph-based orchestration with LangGraph (researcher → planner → builder → reviewer); parallel task execution, swarms, and fault tolerance with merge steps; conversational multi-agent collaboration using AutoGen; comparison of orchestration paradigms; hybrid architectures combining structured control and dynamic agent interaction; deep research workflows with evidence synthesis.	7
Chapter No 4: Learning Agents, Deployment, Safety & Governance: Adaptive and learning agents with reinforcement learning concepts; alignment techniques including RLHF, RLAIF, and DPO; context engineering and dynamic retrieval; self-correcting agents using Reflexion; human-in-the-loop systems; cloud-integrated agents and the agentic SDLC; performance considerations, failure handling, and observability; safety and governance with approval gates, audit logs, and evaluation; real-world applications, engineering capstone, and future directions of autonomous AI systems.	8

Text Books

1. C. Huyen, "AI Engineering: Building Applications with Foundation Models", 1st ed., O'Reilly Media, Dec. 2024.
2. Roberto Infante, "AI Agents and Applications", Manning Publications, 2025.

References

1. David L. Poole and Alan K. Mackworth, "Artificial Intelligence: Foundations of Computational Agents", 2nd Edition, Cambridge University Press, 2017.

Evaluation Scheme (ISA)

SL. No.	Section	Conducted marks	for	Weightage in Marks
i.	ISA -I	30		33
ii.	ISA – II	30		
iii.	Lab Evaluation	30		17
iv.	Course Activity	20		
Total				50

Evaluation Scheme (ESA)

SL. No.	Section	Conducted marks	for	Weightage in Marks
i.	Theory Exam	60		33
ii	ESA Lab	40		17
Total				50

In-Semester Assessment Scheme

Course Unitization for Minor Exams and End Semester Assessment

Topics / Chapters	Teaching Credits	No. of Questions in Minor Exam-1	No. of Questions in Minor Exam-2	No. of Questions in Activity	No. of Questions in ESA
Unit I					
1. Foundations of Agentic AI	7	1.5	-	-	1.5
1. Agent Frameworks, Tools, Skills & Memory	8	1.5	-	-	1.5
Unit II					
Multi-Agent Orchestration	7	-	1.5	-	1.5
Learning Agents, Deployment, Safety & Governance	8	-	1.5	-	1.5

Note

- Each Question carries 15 marks and may consists of sub-questions.
- Mixing of sub-questions from different chapters within a unit (only for Unit I and Unit II) is allowed in Minor I, II and ESA
- Answer 4 full questions of 15 marks each (two full questions from Unit I, II) out of 6 questions in ESA.

Date:

HoD, CSE

Course Assessment Plan

Course Title: Agentic AI				Code: 26ECAC401		
Course outcomes (COs)	Weightage in assessment	Assessment Methods				
		ISA1	ISA2	Lab activity and course project	ESA-Lab exam	ESA-Theory Exam
1. Explain the foundations of agentic AI, including the evolution from Generative AI to Agentic AI, the agent lifecycle, the plan-act-observe-revise loop, and design principles for autonomous, goal-driven agents.	25%	✓		✓		✓
2. Design and implement agentic systems using frameworks, tools, and function calling, reusable skills, and memory management techniques such as the Model Context Protocol (MCP) and agent runtimes.	25%	✓		✓		✓
3. Develop multi-agent orchestration systems using role-based specialization, graph-based coordination with LangGraph, parallel task execution, and conversational multi-agent collaboration with AutoGen.	25%		✓	✓		✓
4. Analyse learning, deployment, safety, and governance considerations for autonomous AI systems, including reinforcement learning-based alignment, self-correcting agents, human-in-the-loop oversight, and the agentic SDLC.	25%		✓	✓		✓
Weightage		25%		25%	50%	

Agentic-AI Lab Plan

List of Experiments

Experiment No.	Brief description about the experiment	Number of slots
1.	Experiments on Planning Agent on Paper and in Python, Plan–Act–Observe–Revise Loop Implementation, Calculator/File Reader/API Caller Tools	3
2.	Experiments on Reusable Research/Code-Review Skill, Short-term State plus Retrieval Store (Memory Management), Mini MCP Connector (Files, DB, API)	3
3.	Experiments on Mini Agent Runtime with Skills and Logs, Researcher → Planner → Builder → Reviewer Graph (LangGraph), Parallel Role Agents with Merge Step	3
4.	Experiments on Evidence Table and Executive Synthesis, Cloud Log Analysis and Remediation Plan, Requirements-to-Test Mini Software Factory, Approval Gates/Audit Logs/Evaluation	3
5.	Course Activity	3

Lab Evaluation Rubrics

Exercise	Criteria	Good (4 M to 5 M)	Average (2 M to 3 M)	Poor (0 M to 1 M)	PI	CO
Planning Agent & Plan-Act-Observe-Revise Loop	Problem Understanding & Objective	Clearly defines the agent's goal, plan, and the observe-revise cycle	Shows general understanding with minor gaps	Poorly defined objectives or lacks clarity	2.1.3	1
	Design of Agent Architecture	Appropriate architecture with clear planning, action, and revision components	Basic architecture design but lacks detail or has some mismatches	Inappropriate or incomplete design of the loop	1.4.1	1
	Implementation	Correct and efficient Python implementation of the planning agent and its loop	Implementation works with minor bugs or limited functionality	Partial implementation or significant issues in agent behavior	5.3.1	1
Tool Integration using LangChain (Calculator, File Reader, API Caller)	Problem Understanding & Use Case Definition	Clearly articulates the objective, scope, and relevance of the tools used	Defines the task and tool usage with some ambiguity	Poor or vague explanation of the problem and tools used	2.1.3	2
	LangChain Tool/Function-Calling Design	Correct use of LangChain tool and function-calling components with a clear pipeline	Basic flow with some components misused or underutilized	Incorrect or incomplete tool design; lacks logical flow	1.4.1	2
	Integration & API Configuration	Seamless integration of tools/APIs; appropriate configuration and error handling	Integration mostly correct; some issues in configuration or output handling	Integration flawed or superficial; outputs poorly handled	5.3.1	2
Memory Management & Mini MCP Connector	Problem Understanding & Use Case Definition	Clearly articulates the objective, scope, and relevance of memory and MCP connectivity	Defines the task with some ambiguity	Poor or vague explanation of the problem and tools used	2.1.3	2

	Memory/Retrieval Store Implementation	Short-term state and retrieval store work effectively and demonstrate context continuity	Runs with minor issues or limited demonstration of memory utility	Implementation incomplete or shows weak understanding of memory use	1.4.1	2
	MCP Connector Design & Output Analysis	Well-designed MCP connector (files/DB/API) with clear, well-analyzed outputs	Reasonable connector design; analysis present but limited in depth	Weak connector design and little or no critical analysis of output	5.3.1	2
Multi-Agent Orchestration using LangGraph/ AutoGen	Problem Understanding & Objective Clarity	Clearly defines agent roles, orchestration graph, coordination mechanisms, and goals	General understanding with minor conceptual gaps	Vague objectives or unclear simulation purpose	2.1.3	3
	Orchestration/Communication Design	Effective use of graph-based orchestration or conversational protocols with realistic interactions	Orchestration logic works but may lack depth or scalability	Minimal or incorrect implementation of orchestration	2.3.1	3
	Simulation Environment & Agent Behavior	Environment supports diverse tasks; agents behave adaptively and demonstrate collective intelligence	Functional environment with some observable behavior	Poor or static environment with minimal observable intelligence	2.3.1	3
	Output Observation & Analysis	Insightful analysis of orchestration flow and task outcomes	Basic observation and comparison; lacks depth in explanation	Superficial or missing analysis; outputs not well discussed	5.3.1	3

Course Activity Evaluation rubrics

Exercise	Criteria	Good (4 M to 5 M)	Average (2 M to 3 M)	Poor (0 M to 1 M)	PI	CO
Course Activity	1. Problem Definition & Agent Roles	Problem well-defined with strong justification for a multi-agent/capstone system; clear, innovative agent roles with autonomy and goals	Problem defined with some justification; agent roles mostly clear	Problem vaguely defined; agent roles unclear or overlapping	1.4.1	4
	2. System Architecture Design	Detailed architecture diagram; clear orchestration and communication design; well-structured coordination & safety strategy	Clear design with some details missing; orchestration/coordination partly explained	Simplistic design with minimal explanation; weak coordination	2.1.3	4
	Implementation (LangChain / LangGraph)	Fully functional prototype; agents integrated with memory, tools, orchestration, and governance controls; smooth workflows	Functional prototype with minor gaps in integration; some agent capabilities missing	Basic implementation ; limited use of frameworks; minimal functionality	5.3.1	4
	Report writing & Documentation	Comprehensive, well-structured capstone report with clear objectives, methodology, results, safety analysis, and references; professional formatting	Report complete with minor clarity/structure issues; good documentation	Report has basic sections but lacks detail, depth, or consistency	10.1.3	4

Chapter wise Plan

Course Code and Title: 26ECAC401 /Agentic AI	
Chapter Number and Title: 1. Foundations of Agentic AI	Planned Hours:7 hrs

Learning Outcomes: -

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Explain the evolution from Generative AI to Agentic AI and the limitations of standalone LLM-based applications.	1	L2	1.4
2. Describe the definition and characteristics of autonomous agents and their lifecycle.	1	L2	2.3
3. Apply the plan–act–observe–revise loop and agent architecture to design goal-driven agents.	1	L3	2.1
4. Apply goal decomposition, iterative reasoning, and planning strategies to build scalable agents.	1	L3	2.3

Lesson Schedule
Class No. - Portion covered per hour
1. Evolution from Generative AI to Agentic AI systems
2. Limitations of standalone LLM-based applications in real-world deployments
3. Definition and characteristics of autonomous agents
4. Agent lifecycle and the plan–act–observe–revise loop
5. Agent architecture
6. Goal decomposition and iterative reasoning
7. Planning strategies and design principles for scalable, goal-driven AI agents

Review Questions

S.NO	Questions	TLOs	BL	PI Code
1	Discuss the evolution from Generative AI to Agentic AI, highlighting the limitations of standalone LLM-based applications in real-world deployments.	1	L3	1.4.1

2	Define an autonomous agent. Explain its key characteristics with real-world examples.	2	L2	2.3.1
3	Explain the plan–act–observe–revise loop with a neat diagram. Why is this loop central to agentic behaviour?	3	L2	1.4.1
4	Describe a generic agent architecture and give a practical example where it is suitable.	3	L2	2.1.3
5	What is goal decomposition? Explain how iterative reasoning helps an agent break down a complex task.	4	L3	1.4.1
6	Compare at least two planning strategies used to design scalable, goal-driven AI agents.	4	L3	2.3.1
7	A customer-support chatbot is traditionally a single-turn, stateless LLM application. If you redesign it as an agentic system, what changes would you introduce to its lifecycle and loop?	2	L3	2.3.1
8	Develop an AI tutor for teaching mathematics using an Agentic AI approach. Explain how you would structure its plan–act–observe–revise loop to adapt to different student learning speeds.	3	L3	2.3.1
9	Discuss the design principles that make an agent architecture scalable and goal-driven, with an example.	4	L3	2.3.1
10	A government plans to deploy autonomous monitoring agents in public infrastructure. Analyze the risks of relying on a standalone LLM application versus a properly designed autonomous agent for this task.	2	L3	2.3.1

Course Code and Title: 26ECAC401 /Agentic AI	
Chapter Number and Title:2. Agent Frameworks, Tools, Skills & Memory	Planned Hours:8 hrs

Learning Outcomes: -
At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Design agentic systems using frameworks such as LangChain.	2	L3	1.4
2. Apply tool integration and function calling, including API orchestration, code execution, and external system interaction.	2	L3	2.1
3. Build reusable skills for agentic systems.	2	L3	1.4
4. Apply memory management using short-term state and long-term retrieval stores.	2	L3	1.4
5. Apply context handling and optimization techniques in agents.	2	L2	1.4
6. Use the Model Context Protocol (MCP) and connectors to link agents with external systems.	2	L3	1.4
7. Design a unified agent architecture and agent runtime patterns (OpenCLA/OpenClaw) with logging for real-world deployment.	2	L4	2.3

Lesson Schedule
Class No. - Portion covered per hour
1. Design of agentic systems using LangChain – overview and use cases
2. Tool integration and function calling – API orchestration, code execution
3. External system interaction and building reusable skills
4. Memory management – short-term state
5. Memory management – long-term retrieval stores
6. Context handling and optimization
7. Model Context Protocol (MCP) and connectors
8. Unified agent architecture and agent runtime patterns (OpenCLA/OpenClaw) with logging

Review Questions

Sl.No.	Questions	TLOs	BL	PI Code
1	What is LangChain? Describe how it is used to design agentic systems.	1	L2	1.4.1
2	Explain function calling and API orchestration in the context of tool-using agents.	2	L2	2.3.1
3	Suppose you are building a legal document analysis assistant. How would you design its tool integration (retrieval, summarization, reasoning) so that it behaves more agentic rather than a passive question-answering system?	2	L3	2.1.4
4	What is a reusable skill in an agentic system? Give an example of a research or code-review skill.	3	L2	2.3.1
5	Differentiate between short-term state and long-term retrieval store in agent memory.	4	L2	2.1.3
6	Explain how context handling and optimization improve the performance of an LLM-based agent.	5	L2	1.4.1
7	What is the Model Context Protocol (MCP)? Explain how MCP connectors extend an agent's access to files, databases, and APIs.	6	L3	1.4.1
8	Describe a unified agent architecture that combines reasoning, memory, tools, and environment. Support your answer with a neat diagram.	7	L3	2.3.1
9	For a career-counseling agent that adapts to a student's history. How should the agent use memory and reusable skills to provide long-term personalized guidance?	4	L3	2.1.3
10	Explain the role of agent runtime patterns (OpenCLA/OpenClaw) and logging in deploying agentic systems for real-world use.	7	L3	2.3.1

Course Code and Title: 26ECAC401 /Agentic AI	
Chapter Number and Title:3. Multi-Agent Orchestration	Planned Hours:7 hrs

Learning Outcomes:

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Design distributed AI systems using multi-agent architectures with role-based specialization and coordination.	3	L3	2.3
2. Apply agentic node graphs and graph-based orchestration with LangGraph (researcher → planner → builder → reviewer).	3	L3	2.1
3. Apply parallel task execution, swarms, and fault tolerance with merge steps in multi-agent systems.	3	L3	2.3
4. Design conversational multi-agent collaboration using AutoGen and compare orchestration paradigms.	3	L3	2.1
5. Develop deep research workflows with evidence synthesis using hybrid orchestration architectures.	3	L3	2.3

Lesson Schedule
Class No. - Portion covered per hour
1. Designing distributed AI systems using multi-agent architectures
2. Role-based specialization and coordination strategies
3. Agentic node graphs and graph-based orchestration with LangGraph
4. Parallel task execution, swarms, and fault tolerance with merge steps
5. Conversational multi-agent collaboration using AutoGen
6. Comparison of orchestration paradigms; hybrid architectures
7. Deep research workflows with evidence synthesis

Review Questions

Sl.No.	Questions	TLOs	BL	PI Code
1	Explain how a multi-agent architecture uses role-based specialization for distributed problem-solving.	1	L2	2.3.1

2	Design a legal-research assistant as a researcher → planner → builder → reviewer graph using LangGraph. Compare it with directly chaining prompts.	2	L3	2.3.1
3	A multi-agent customer-support system works alongside human agents. Give an example where the agent graph leads the interaction and another where a human must intervene. How would you balance trust and control?	4	L3	2.3.1
4	What challenges arise in coordinating parallel task execution and merge steps, and how can fault tolerance be achieved?	3	L2	2.3.1
5	How does conversational multi-agent collaboration using AutoGen differ from graph-based orchestration with LangGraph?	4	L3	2.1.3
6	Compare centralized and decentralized orchestration paradigms for multi-agent systems.	4	L2	2.1.3
7	Explain deep research workflows with evidence synthesis. How do parallel role agents contribute to the final synthesis?	5	L3	2.3.1
8	Design a hybrid architecture combining structured control and dynamic agent interaction for a disaster-response drone swarm.	4	L3	2.1.3
9	Suppose you are developing a financial-advisory multi-agent system. How would you design a researcher → planner → builder → reviewer graph for: (a) extracting spending patterns, (b) analyzing investment risks, (c) suggesting portfolio strategies?	2	L3	2.3.1
9	Suppose you are developing a financial advisory agent. How would you design a prompt chain for the following workflow: (a) extract spending patterns, (b) analyze investment risks, (c) suggest portfolio strategies?	5	L3	2.3.1

Course Code and Title:26ECAC401/Agentic AI	
Chapter Number and Title:4. Learning Agents, Deployment, Safety & Governance	Planned Hours:8 hrs

Learning Outcomes:
At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
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1. Explain adaptive and learning agents using reinforcement learning concepts and alignment techniques (RLHF, RLAIF, DPO).	4	L2	1.4
2. Apply context engineering, dynamic retrieval, and self-correcting agents using Reflexion.	4	L3	1.4
3. Analyze human-in-the-loop systems, cloud-integrated agents, and the agentic SDLC for deployment.	4	L4	2.3
4. Analyze safety and governance practices – approval gates, audit logs, evaluation – and real-world applications of autonomous AI systems.	4	L4	2.1

Lesson Schedule
Class No. - Portion covered per hour
1. 1. Adaptive and learning agents – reinforcement learning concepts
2. 2. Alignment techniques: RLHF, RLAIF, and DPO
3. 3. Context engineering and dynamic retrieval
4. 4. Self-correcting agents using Reflexion
5. 5. Human-in-the-loop systems
6. 6. Cloud-integrated agents and the agentic SDLC
7. 7. Performance considerations, failure handling, and observability
8. 8. Safety and governance: approval gates, audit logs, evaluation; real-world applications and future directions

Review Questions

Sl.No.	Questions	TLOs	BL	PI Code
1	Explain how reinforcement learning enables adaptive and learning behaviour in agents.	1	L2	1.4.1
2	Compare RLHF, RLAIF, and DPO as alignment techniques for learning agents.	1	L3	1.4.1
3	What is context engineering? Explain how dynamic retrieval improves an agent's responses.	2	L2	2.3.1
4	Describe how a Reflexion-based agent self-corrects after making an incorrect prediction, using a healthcare diagnostic example.	2	L3	2.3.1

5	Discuss the role of human-in-the-loop systems in safe agent deployment.	3	L2	2.3.1
6	Explain cloud-integrated agents and the agentic SDLC with a neat diagram.	3	L3	2.1.4
7	What performance and observability considerations matter when handling agent failures in production?	3	L2	2.1.3
8	A government plans to deploy autonomous agents in public infrastructure. Discuss how approval gates, audit logs, and evaluation support safe governance, and analyze the risk-efficiency trade-off.	4	L3	2.3.1

Model Question Paper for Minor Examination (ISA-I)						
Course Code: 26ECAC401		Course Title: Agentic AI				
Duration: 60 mins		Max. Marks: 30				
Note: Answer any two full questions. Each full question carries equal marks.						
Sl.No.	Questions	Marks	CO	BL	PO	PI Code
1a	Smart agriculture systems are moving from standalone LLM-based advisory apps to autonomous agents. Using the plan–act–observe–revise loop, describe how you would structure such an agent to support farmers in real-time decision-making.	10	CO1	L3	2	2.1.3
1b	What are the main limitations of standalone LLM-based applications that motivate the shift to agentic AI?	5	CO1	L2	1	1.4.1
2a	Design a LangChain-based agent for a personalized e-learning tutor, showing how tool integration and reusable skills would impact the agent’s performance.	10	CO2	L3	2	2.1.3
2b	Explain the role of goal decomposition in AI agents and how it helps align an agent’s decisions with long-term objectives.	5	CO1	L2	1	1.4.1
3a	Consider a disaster-response drone. Apply the concept of a unified agent architecture and describe how memory, tools, reasoning, and environment components function together.	10	CO2	L3	2	2.1.3
3b	Explain how the Model Context Protocol (MCP) enhances the capability of LLM agents. Why is a retrieval store important for sustaining meaningful interactions?	5	CO2	L2	1	1.4.1

Model Question Paper for Minor Examination (ISA-II)						
Course Code: 26ECAC401		Course Title: Agentic AI				
Duration: 60 mins		Max. Marks: 30				
Note: Answer any two full questions. Each full question carries equal marks.						
Q. No.	Questions	Marks	CO	BL	PO	PI Code
1a	A multi-agent research assistant works alongside human analysts. Give an example of a situation where the researcher → planner → builder → reviewer graph leads the workflow and another where a human must intervene. How would you balance trust and control?	10	CO3	L3	2	2.1.3
1b	Explain the dynamics of conversational multi-agent collaboration using AutoGen and the factors that influence effective teamwork among agents.	5	CO3	L2	1	1.4.1
2a	Design a coordination mechanism for multiple delivery drones in a smart-city scenario using parallel task execution and merge steps, ensuring efficient resource use by considering: battery level (BL), delivery distance (DD), and payload capacity (PL).	10	CO3	L3	2	2.1.3
2b	Describe how RLHF and RLAIF support alignment of learning agents.	5	CO4	L2	1	1.4.1
3a	Apply Reflexion-based self-correction in an edge-deployed agent scenario and explain how the agentic SDLC supports safe, iterative deployment.	10	CO4	L3	2	2.1.3
3b	How do approval gates and audit logs support governance in autonomous AI systems?	5	CO4	L2	1	1.4.1

Question Paper for End Semester Assessment (ESA)

Course Code: 26ECAC401	Course Title: Agentic AI
Duration: 60 mins	Max. Marks: 60

Note: Answer any 2 full questions from each UNIT I and Unit II

UNIT – I

Q. No	Questions	Marks	CO	PI Code	PO	BL
1a	Smart education systems are shifting from standalone LLM apps to agentic AI. Using the plan–act–observe–revise loop, describe how autonomy and goal-orientation can be structured into such a system to support adaptive teaching and real-time feedback.	10	CO1	1.4.1	1	L3
1b	What are the main differences between a standalone LLM-based application and an agentic system?	5	CO1	1.4.1	1	L2
2a	Design an agent architecture for a personalized e-learning tutor. Explain how tool integration, reusable skills, and memory would impact the agent’s performance.	10	CO2	2.4.1	2	L3
2b	Explain the role of the Model Context Protocol (MCP) in agentic systems and how it helps agents access external context.	5	CO2	2.1.3	2	L2
3a	Consider a disaster-response drone. Apply the concept of a unified agent architecture and describe how perception, reasoning, memory, and action components function together.	10	CO1	2.1.3	2	L3
3b	Explain how short-term state and long-term retrieval stores enhance the performance of LLM agents. Why is memory important for sustaining meaningful interactions?	5	CO2	1.4.1	1	L2
4a	Multi-agent orchestration is crucial in decision-making. Illustrate with an example how a researcher → planner → builder → reviewer graph can assist a financial advisor in investment planning.	10	CO3	2.3.1	2	L3
4b	Define context engineering. How does it enhance the reasoning ability of LLM-based agents?	5	CO4	1.4.1	1	L2
5a	In a multi-agent logistics system, explain how parallel task execution and merge steps are achieved among delivery drones considering limited resources and dynamic conditions.	10	CO3	2.1.3	2	L3
5b	Discuss how RLHF and Reflexion support adaptive, self-correcting behavior in learning agents.	5	CO4	2.3.1	2	L2
6a	Agents often rely on conversational protocols in multi-agent orchestration. Design a communication strategy for autonomous vehicles in a smart-city scenario using AutoGen and explain how it ensures road safety.	10	CO4	2.1.3	2	L3

6b	How can approval gates and audit logs help autonomous AI systems adapt their governance dynamically during safety incidents?	5	CO3	2.1.3	2	L2
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